

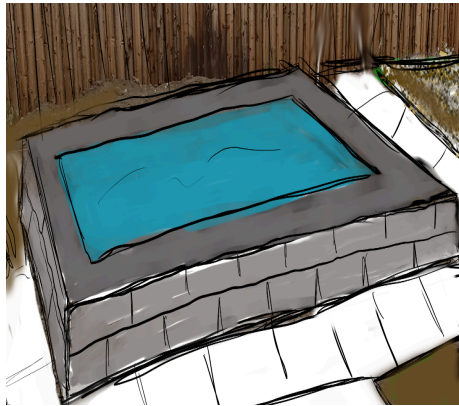
'12 x '16 x '4 CMU WALL SWIMMING POOL

Proposed construction under a Homeowner-Affidavit permit; of a swimming pool utilizing Concrete Masonry Unit (CMU) wall construction set upon a reinforced concrete slab. The CMUs, will be partially earth-retained, are to receive a multi-layered finish system with below-grade waterproofing, cementitious plaster, and a durable, pool-compatible coating. A filtration and sanitation system will be integrated, and designed to current industry standards. Hydrostatic pressure relief and groundwater management measures are included. A separate 5' x 3' concrete equipment pad, is proposed approximately '12 from the primary pool structure to house essential plumbing and mechanical components. All subterranean plumbing shall be placed within properly excavated trenches, under frost line, backfilled with free-draining material and a sand layer, strictly adhering to all relevant codes. All reinforcing steel will maintain 6" clearance from earth and formwork prior to pour, and will have a earth bonding system. Two faces of the main structure are to have a concrete walkway that is to be embedded into the CMUs with epoxy. Project is to use existing electrical that is located in proximity to proposed equipment pad and $\geq$ '12 from proposed swimming pool.

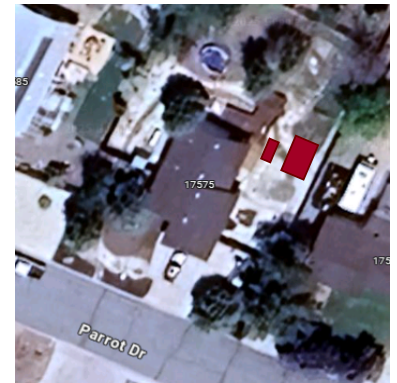
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Artistic



Example



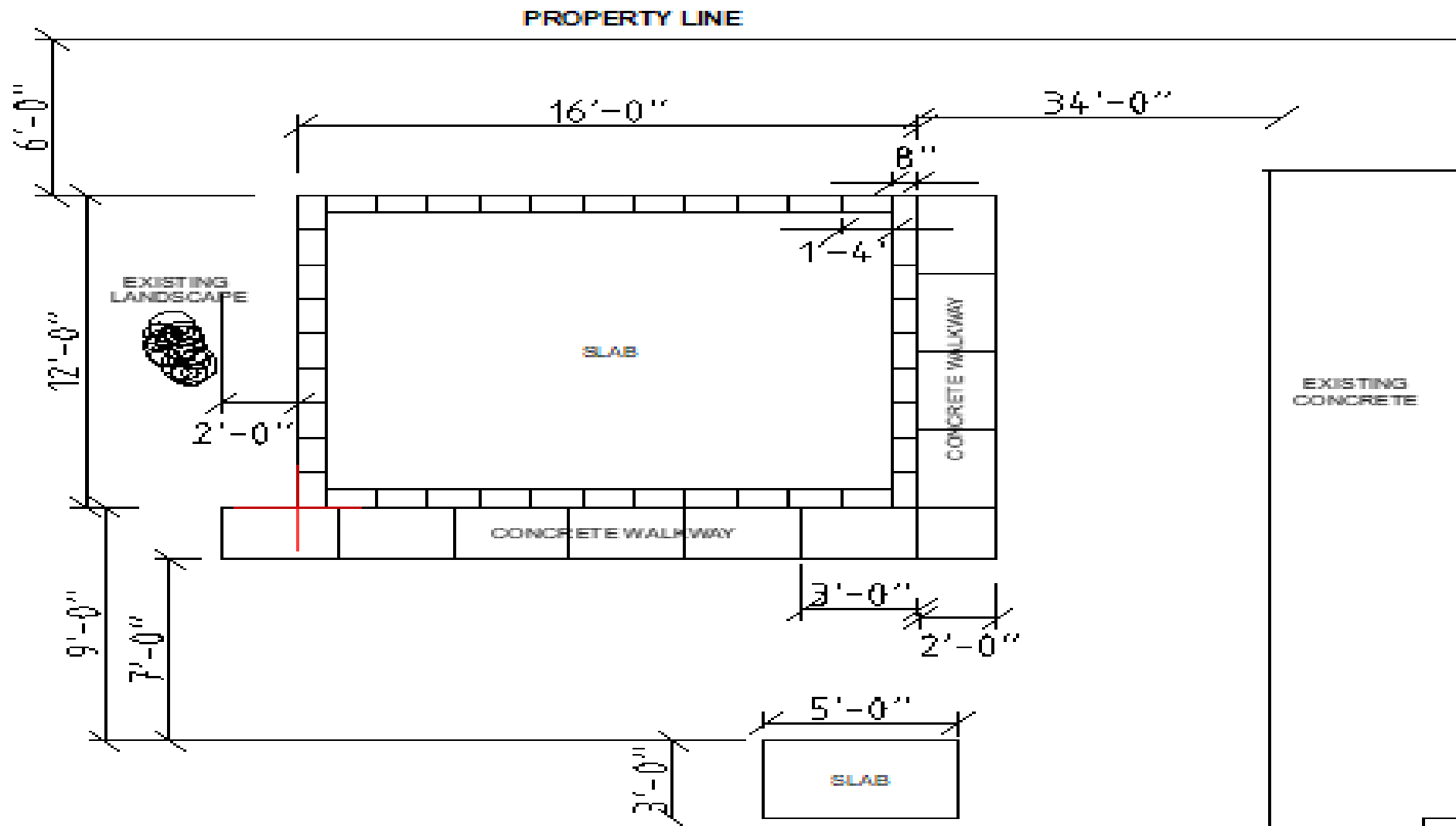
Approximate proposed location

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Cover
G1.01

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SITE PLAN

- * '12 x '16 x 6" MAIN SLAB (BELOW GRADE)
- * '5 x '3 x 6" SECONDARY SLAB (AT GRADE)
- * CONCRETE WALKING PATH
ALONG SOUTH/WEST FACES
WITH '3 x '2 x 4" FLAGS
- * '6 PROPERTY LINE SETBACK
- * CMU WALLS ON MAIN SLAB
- * CMU DIMENSIONS: 8" x 8" x 16"
- * 4" PERFORATED PVC DRAIN UNDER MAIN SLAB
- * #4 REBAR FOR BOTH SLABS
- * MAIN SLAB HAS COMPACTED FILL AND VAPOR/MOISTURE
BARRIER INSTALLED ON TOP OF DRAINAGE TRENCH/PIPE
- * NOT TO SCALE *

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Structural Overview

Structural 3.01

1 – Water volume

Volume (cu ft) = length × width × depth = 16 × 12 × 4.

Total gallons = 5,745.03936

2 – Concrete volume

Footing

Perimeter length = 56 ft.

Footing cross-section = 16" × 8" = (16/12) ft × (8/12) ft.

Footing total volume = 1.844 yd³

Main Slab

Area = 12 × 16 = 192 sq ft.

Thickness = 6" = 0.5 ft.

Volume = area × thickness = 192 × 0.5 = 96 cu ft.

Slab total volume = 3.556 yd³.

3 – Slab Grid

Slab reinforcement: #4 bars @ 8" o.c. both directions (placed mid-depth).

Number of bars/ft long-direction (parallel to long side) = 208 ft.

Number of bars/ft short-direction = 204 ft.

Total linear feet of slab #4 = 208 + 204 = 412 ft.

4 – Summary

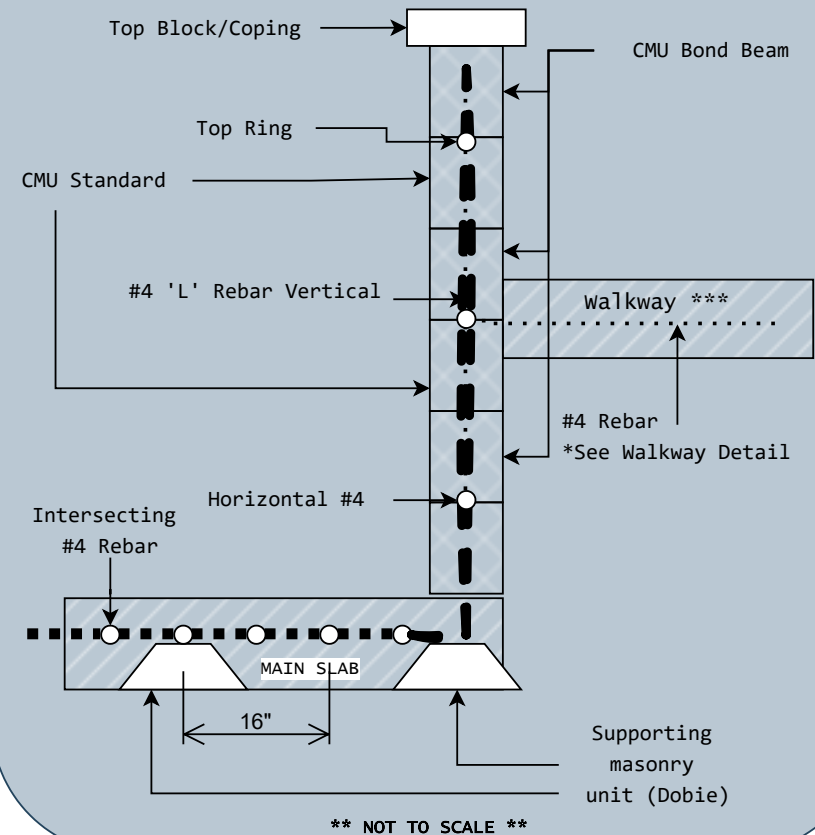
Footing: 2 #4 continuous (perimeter) – total ≈ 112 ft.

Bond beam: 1 #4 horizontal continuous – 56 ft (specify if 2 required).

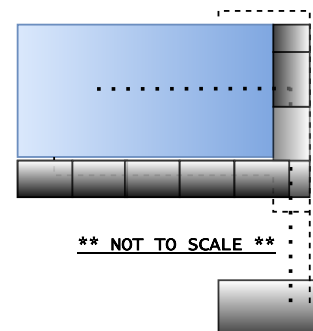
Vertical wall: #4 @ 24" o.c. – 28 bars, total ≈ 149.33 ft.

Slab: #4 @ 12" o.c. both directions – total ≈ 412 ft.

Cross Section Overview



Overview

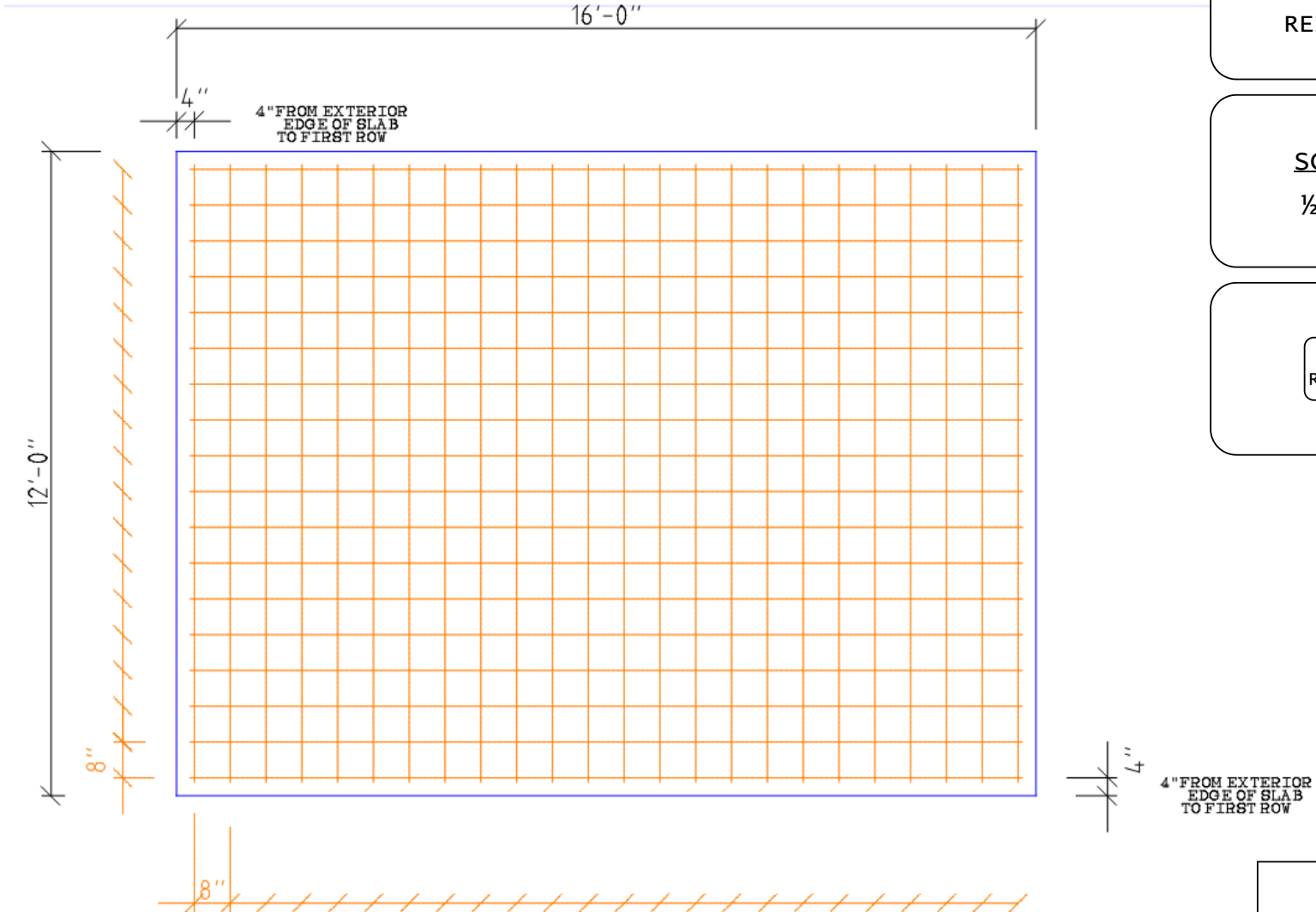


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Structural
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Structural 3.02

MAIN SLAB
REINFORCEMENT

SCALE: 1/2

1/2" = 1'-0"

#4 1/2"
Reinforcement

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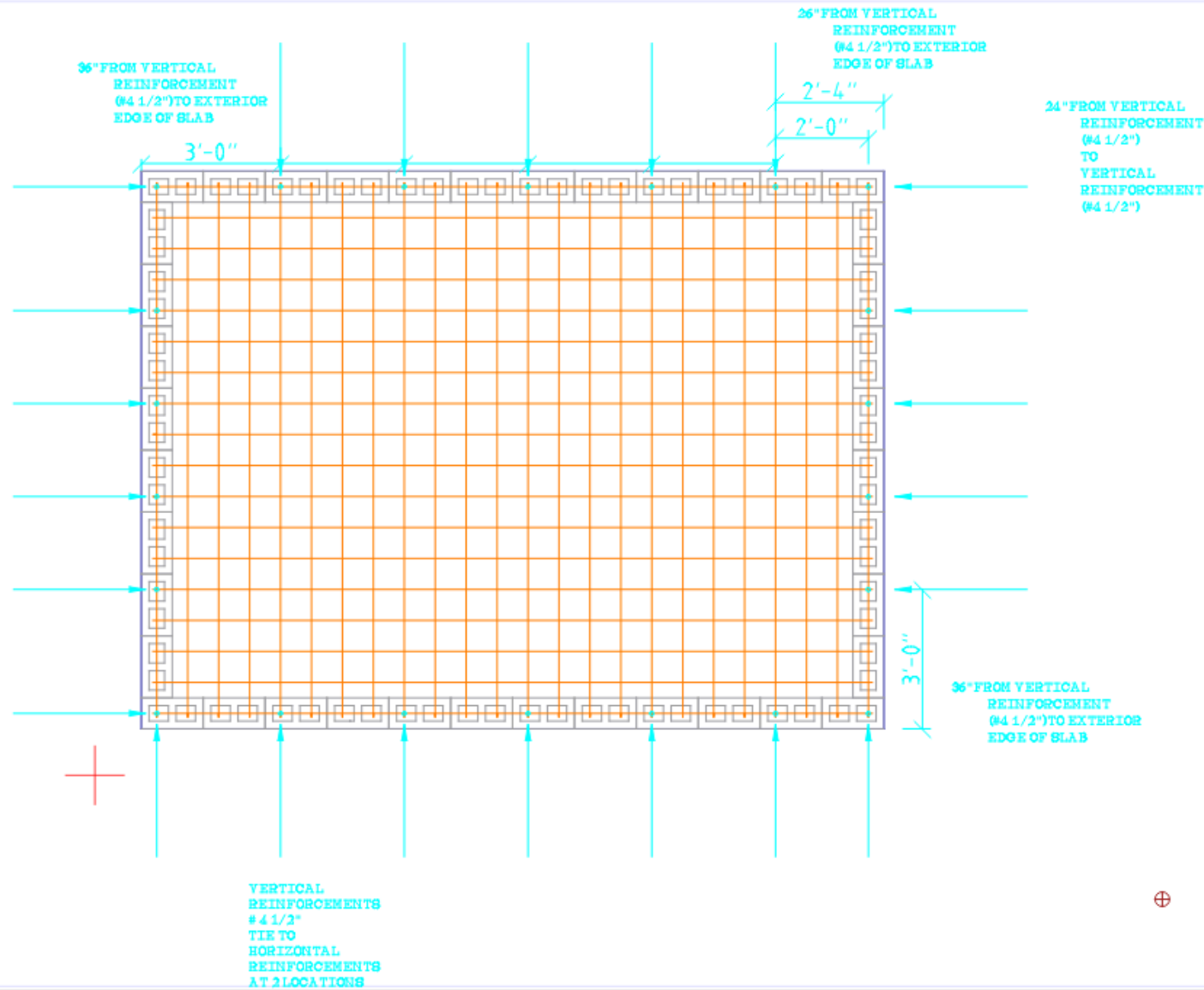
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Structural 3.03

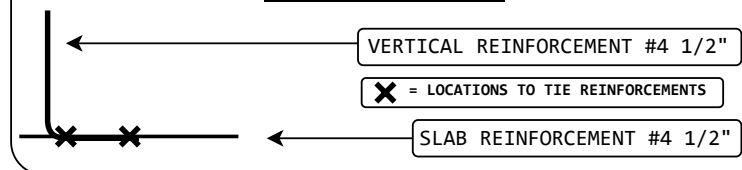
MAIN SLAB VERTICAL REINFORCEMENT

SCALE: 1/36

1in = 3ft



CROSS SECTION DETAIL



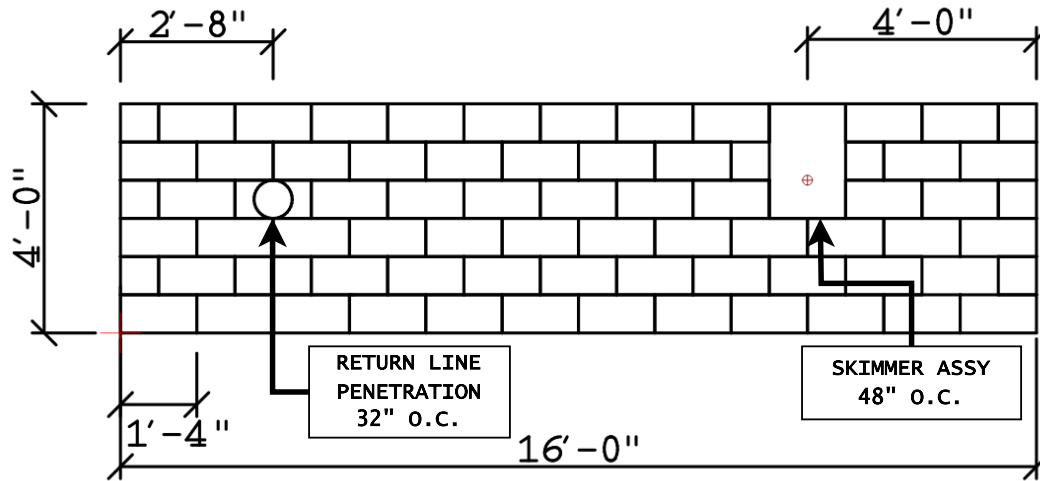
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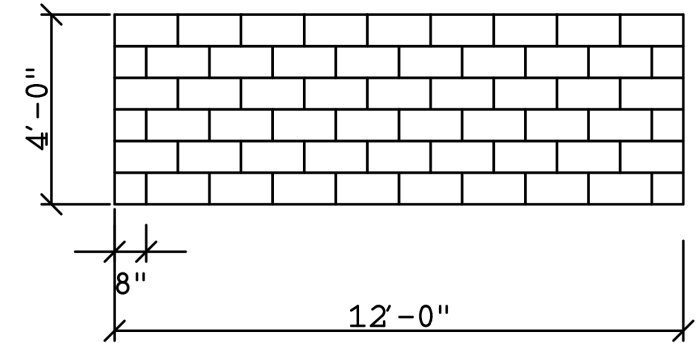
Structural
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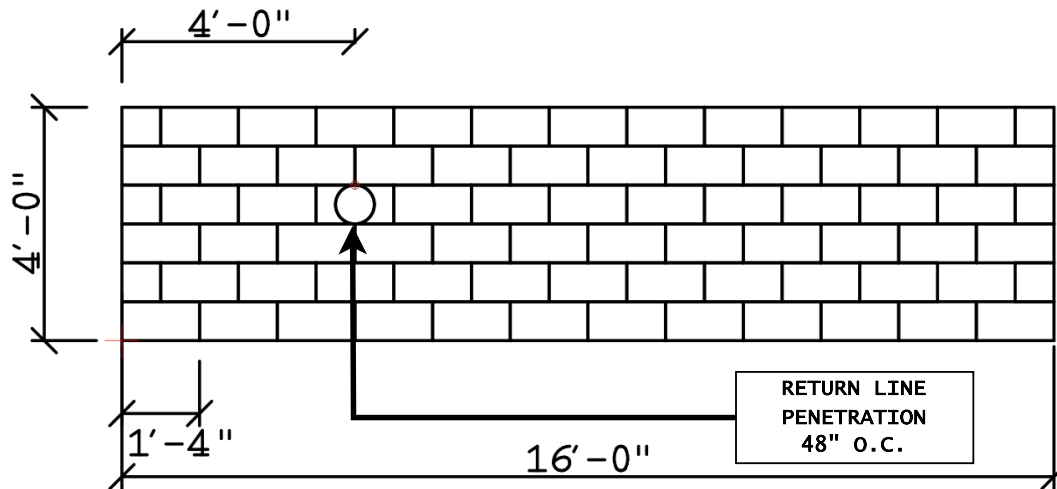
Block Layout (East Elevation)



Block Layout (North & South Elevations)



Block Layout (West Elevation)



Structural 3.04

- * Standard CMU for first course, third course, and fifth course
- * Bond beam CMU for second course, fourth course, and sixth course
- * Skimmer assembly to be 48" from right edge on East Elevation
- * Return line on East Elevation is 32" O.C. from left edge
- * Return line on West Elevation is 48" O.C. from left edge
- * Vibrated concrete for CMU cells on all vertical walls

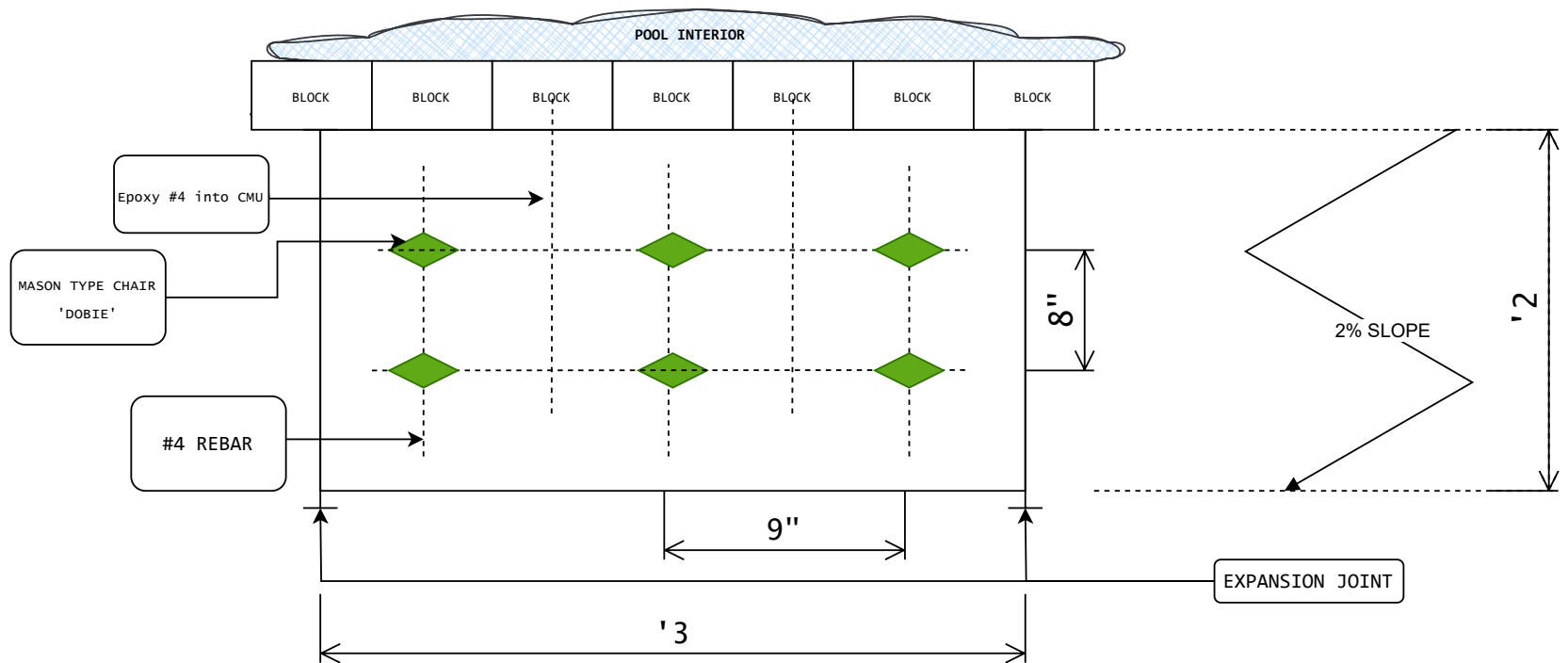
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WALKWAY DETAIL (SINGLE FLAG)



Structural 3.12

- * All slabs to use #4 Rebar
- * Repeat this detail for every flag

**** NOT TO SCALE ****

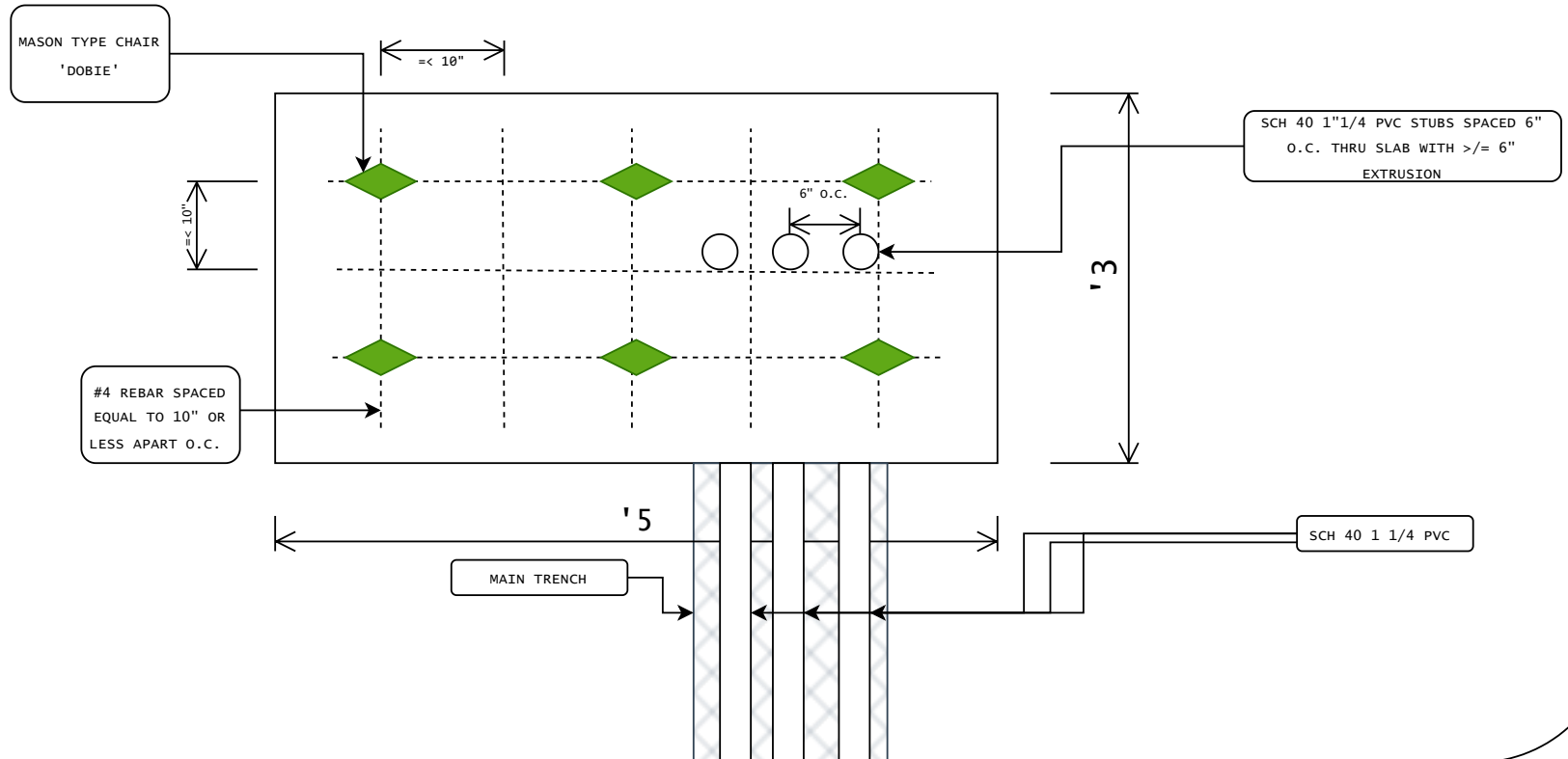
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SECONDARY SLAB (Pump Pad)



Structural 3.11

- * All slabs to use #4 Rebar
- * Seperate plumbing stubs 6" O.C.

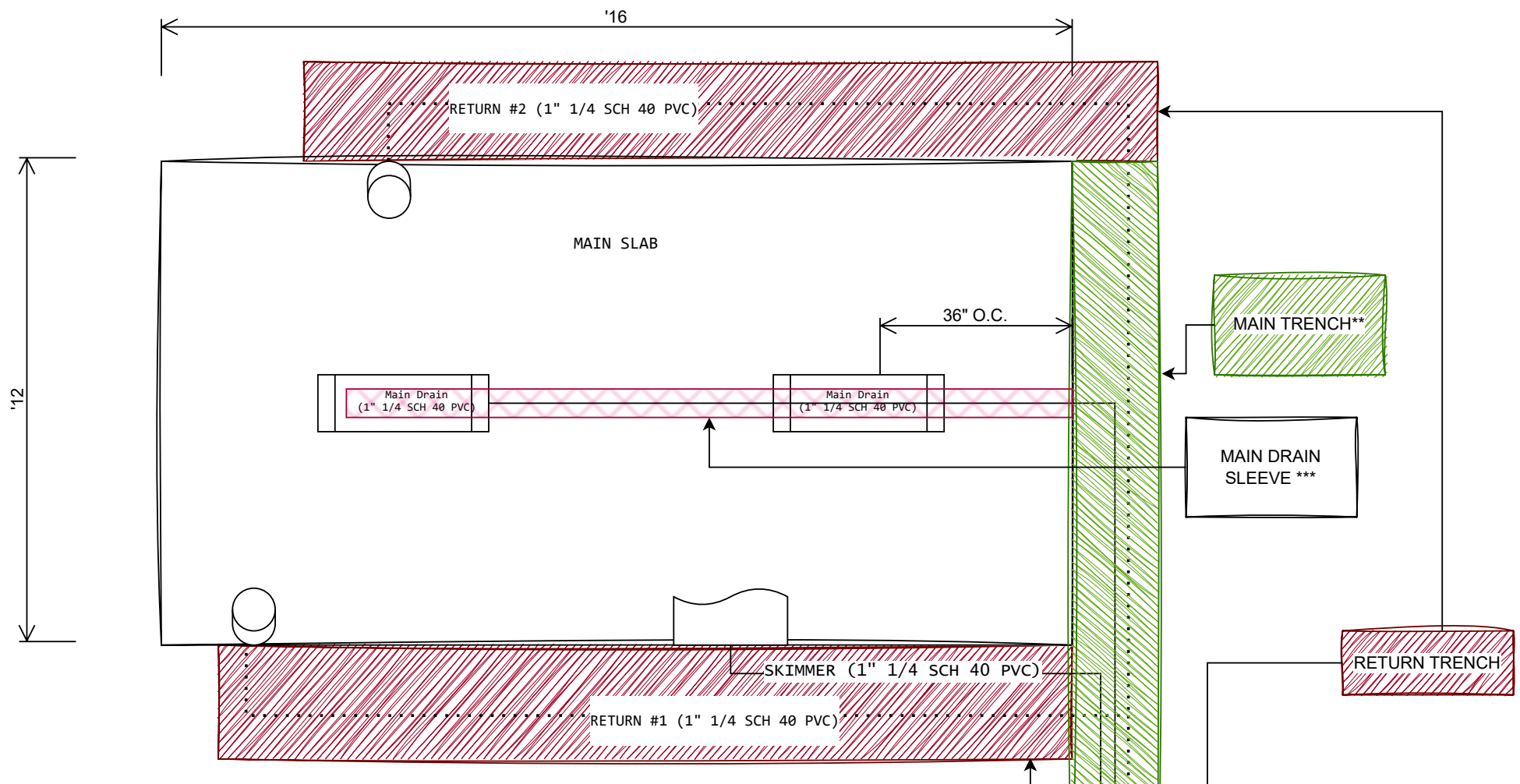
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Structural
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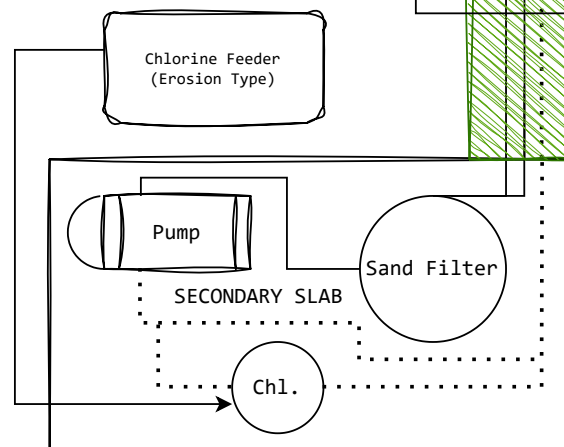
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Plumbing Overview 1

- Solid Line denotes Suction Side of the system
- Dotted Line denotes Pressure Side of the system
- * System to be piped in SCH 40 PVC 1" 1/4 in compliance with local code requirements.
- ** Trenches are to be below frostline, with a base of permeabl fill then topped with sand.
- *** Portion of Main Drain line under slab to be sleeved inside of 4" Perforated PVC for hydrostatic relief and water table changes

**** NOT TO SCALE ****

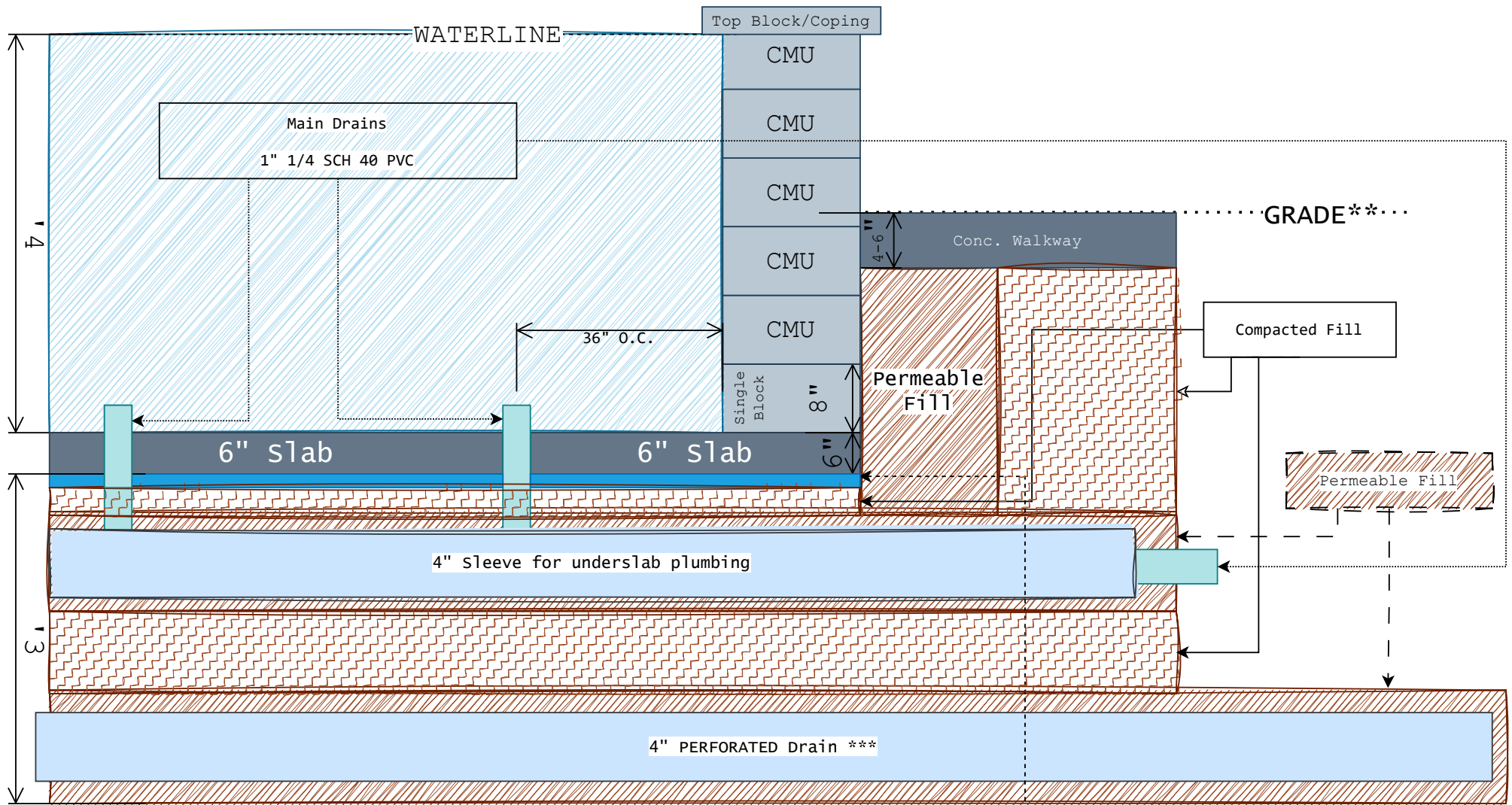


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PLUMB
P1.01

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Overview: Cross Section (East Elevation)

* Main Drain piping to run inside 4" sleeve under slab to trench

** Grade will vary

*** 4" drain to run under slab for hydrostatic relief, saturated earth and site drainage (See Geo/Civil)

**** Minimum 10 mil sheeting

(Not To Scale)

Vapor/Moisture Underlayment ****

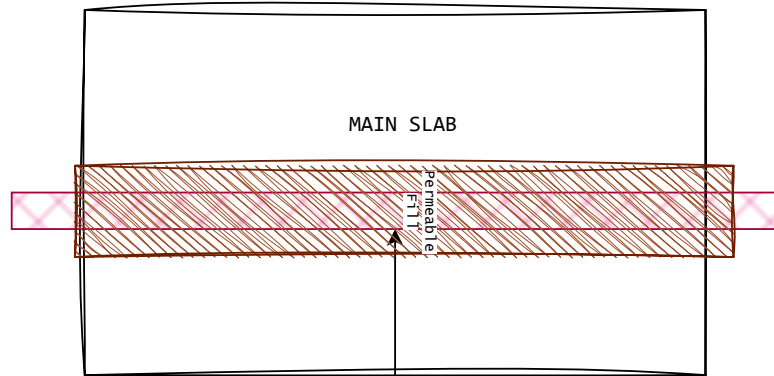
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General
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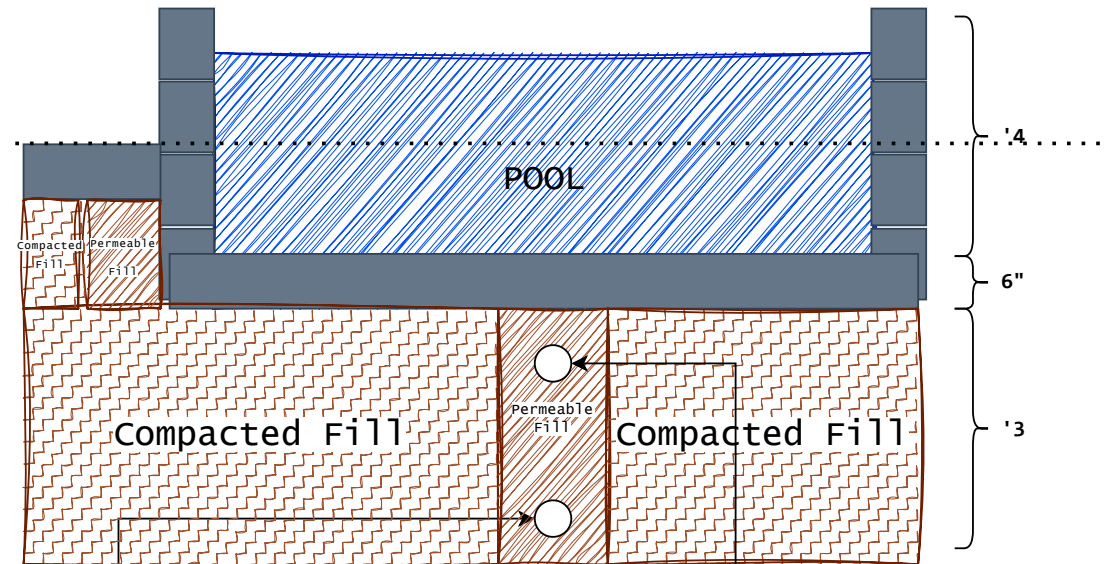
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OVERVIEW



Perforated 4" PVC Drain
* "FRENCH DRAIN" for ground water

CROSS SECTION



Main Drain Trench

Main Drain for
filtration system

Geotechnical / Civil (Drainage)

- * 4" Perforated PVC Drain to run under entire main slab for groundwater control
- * Surround Drain with Permeable Fill for hydrostatic relief

**** NOT TO SCALE ****

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Geo/Cvl
B/C 1.01

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